

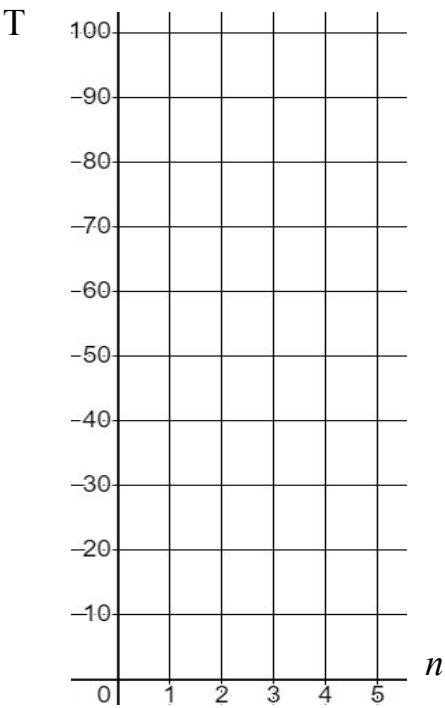


Representing geometric sequences graphically

Task A: Graphing a common ratio greater than 1

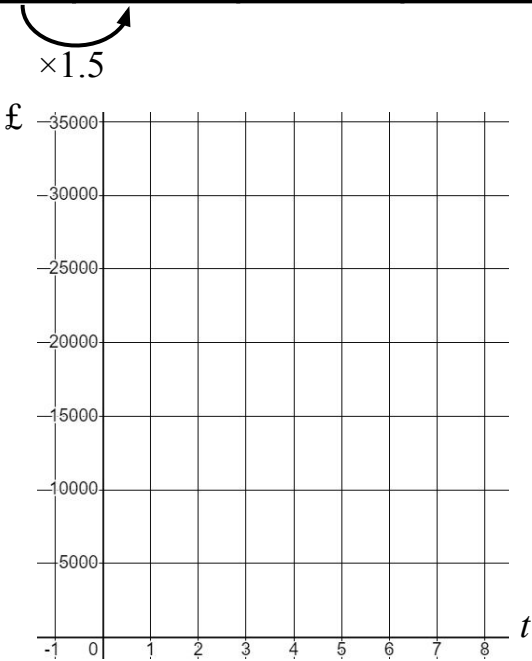
1) Fill in the table of values and plot the geometric sequence that starts on 1 and has a common ratio of 3

Term number (n)	1	2	3	4	5
Term value (T)					



2) You invest £2000 in a new tech company and your shares grow in value by 50% each year. Calculate their value for the first 8 years and plot it on the graph.

t	1	2	3	4	5	6	7	8
£	2000							



Hint: To increase by 50%
is a multiplier of 1.5
Hint: Round your values to the
nearest £1 and plot approximately.

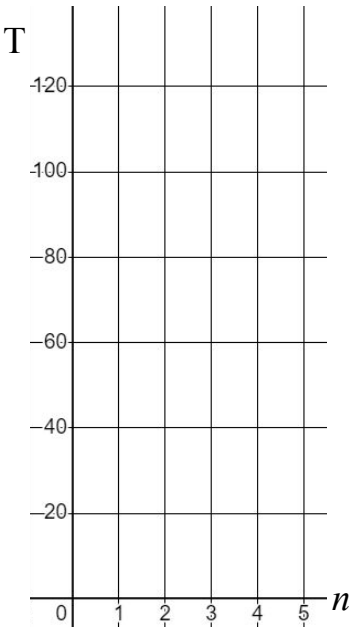
3) Sam is contemplating the shape of the graphs of geometric sequences.



I don't know if the starting term affects the shape. If I use the same common ratio of 2 but start on two different numbers will I get a different shape?

<i>n</i>	1	2	3	4	5
T	2				

<i>n</i>	1	2	3	4	5
T	8				



Using a common ratio to complete the tables, plot both graphs, and write a conclusion for Sam.

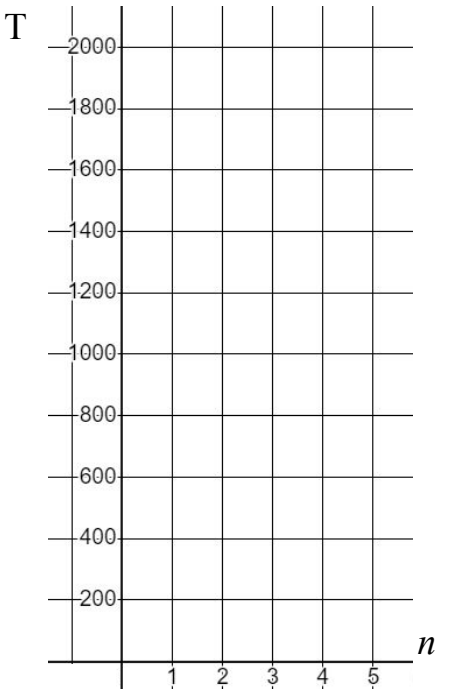
Beware, you can only plot the points approximately.

Task B: Graphing a common ratio less than 1

1) a) Complete the table of values and graph this geometric sequence with a common ratio $\frac{1}{3}$

<i>n</i>	1	2	3	4	5
T	2025				

$\times \frac{1}{3}$



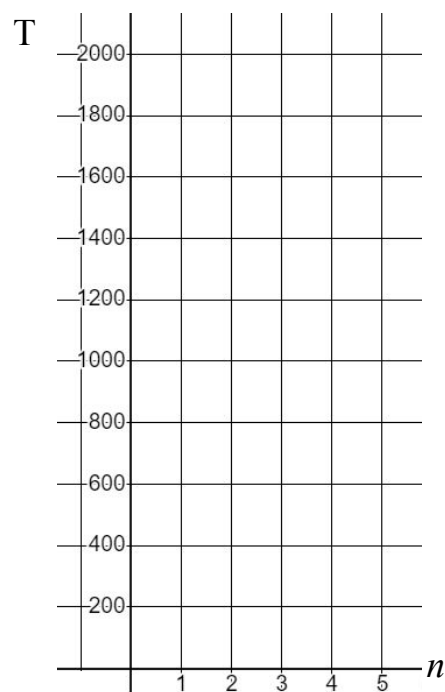
Plot the coordinates as accurately as you can.

1) b) Complete the table of values and graph this geometric sequence with a common ratio $\frac{2}{3}$

n	1	2	3	4	5
T	2025				

$\times \frac{2}{3}$

Plot the coordinates as accurately as you can.



2) Complete the table of values and graph this geometric sequence with a common ratio -1.5

n	1	2	3	4	5	6	7	8
T	2.56							

$\times -1.5$

Plot the coordinates as accurately as you can.

